

EJ-200/204/230 Bar: Timing Optimisation Offline Analysis of Existing Geant4 Data

Simulation analysis

2026-08-16

Available ROOT files (all 12 branches)

Dataset	Files	Readout	ph/ev/end
scan_end_vikuiti	39	END	350–1750
scan_resolution	52	END+TOP	1–2 / 107
scan_end_tir	26	END	≈ 1

Key branches: event_id, face_type, global_id, local_id, time_ns, gun_x_mm

What is computable offline:

- k -th order statistic: any $k \leq N_{\text{pe}}$
- Mean of first m photons per end
- Software SiPM ablation via local_id mask
- MC-truth bias via gun_x_mm

Analysis phases:

Phase A k -scan: $T_0^{(k)} = \frac{1}{2}[t_L^{(k)} + t_R^{(k)}]$, $k = 1 \dots 200$

Phase B m -average: $T_0^{(m)} = \frac{1}{2}[\bar{t}_L^{(1:m)} + \bar{t}_R^{(1:m)}]$

Phase C TOP ablation: N_{top} from 70 to 1 (8 random seeds)

Phase D END ablation: N_{end} from 8 to 1 per face (8 random seeds)

Phase E Pareto frontier: $\sigma(T_0)$ vs SiPM count

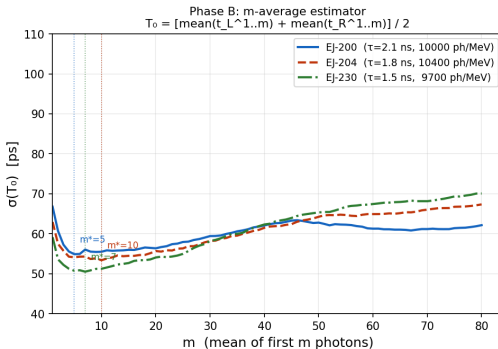
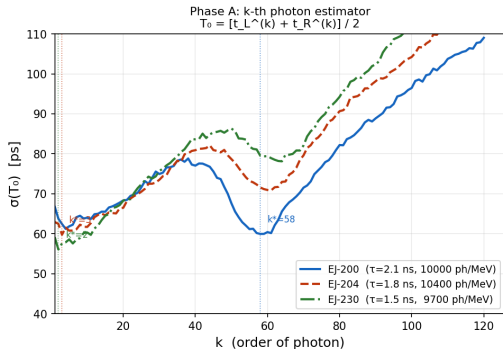
No new Geant4 runs required for these.

All results use IQR-based σ (robust to non-Gaussian tails).

EJ-230: 13 positions $\times$ 5000 ev; N_{pe}/end : 350–1540.

Phase A+B: optimal photon estimator at $x = 0$

END+Vikuiti timing resolution vs estimator — $x = 0$ mm



Key finding: m -average is **systematically better** than k -th single photon (≈ 8 –11% gain in σ).

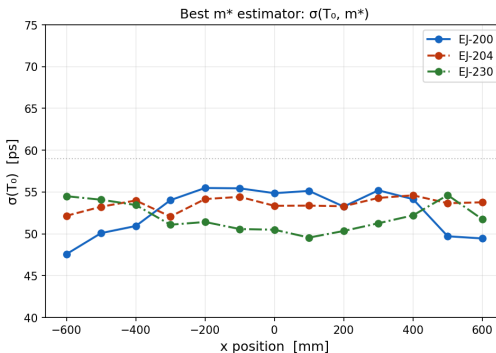
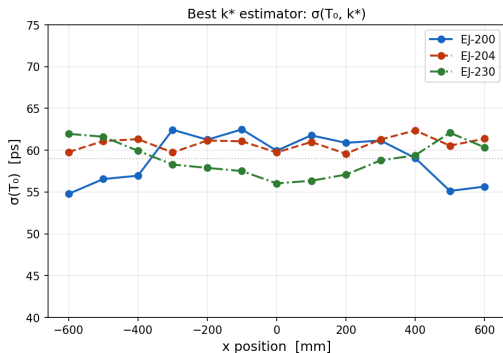
Optimal m^* is small (5–12): *use mean of first few photons, not all.*

Scint.	k^*	$\sigma(k^*)$	$\sigma(m^*)$
EJ-200	58	59.9 ps	54.9 ps
EJ-204	3	59.7 ps	53.4 ps
EJ-230	2	56.0 ps	50.5 ps

EJ-200's large $k^* = 58$ reflects slow $\tau = 2.1$ ns: early photons are noisy. m -average sidesteps this efficiently.

Phase A+B: position dependence

END+Vikuiti: timing resolution vs muon position



EJ-230 best at centre ($x \approx 0$): shorter τ helps when N_{pe} is high.

EJ-200 recovers at extremes ($|x| \sim 600$ mm): longer attenuation ($L_{att} = 380$ cm) keeps photon count high at distant end.

N_{pe} at left end ($x = -600$ mm):

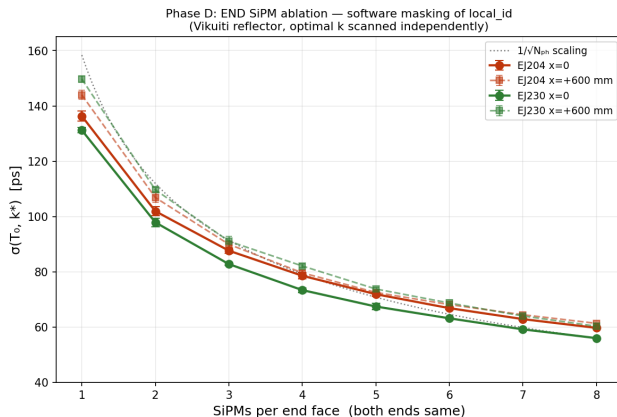
EJ-200: 1765, EJ-204: 1750, EJ-230: 1543.

N_{pe} at left end ($x = 0$ mm):

EJ-200: 1311, EJ-204: 941, EJ-230: 701.

EJ-230's shorter $L_{att} = 120$ cm reduces yield strongly toward centre.

Phase D: END SiPM ablation



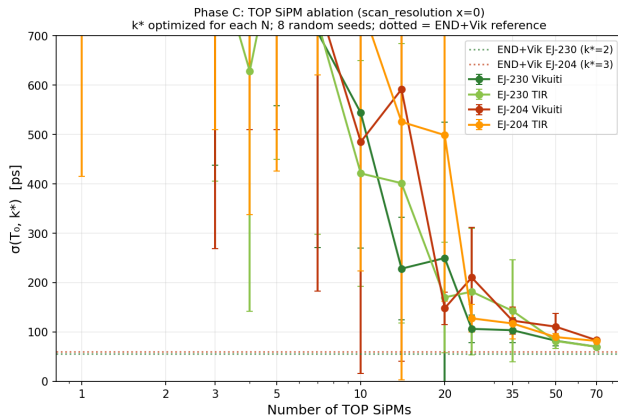
Method: mask local_id $\subset \{0 \dots 7\}$
(same subset for both ends; 8 random seeds).

N/end	N _{tot}	EJ-204	EJ-230
8	16	59.7 ps	56.0 ps
6	12	66.9 ps	63.2 ps
4	8	78.6 ps	73.4 ps
2	4	101.9 ps	97.9 ps
1	2	136.3 ps	131.2 ps

4 SiPMs total (2/end) already gives
< 100 ps for both scintillators.

Scaling: $\sigma \propto N_{pe}^{-0.55}$ (slower than $1/\sqrt{N}$ due to
order-stat structure).

Phase C: TOP SiPM ablation



Method: random subset of local_id from the 70 TOP SiPMs; 8 seeds.

N_{top}	EJ-204 Vik	EJ-230 Vik
70	83.7 ps	70.4 ps
50	111 ps	82.5 ps
35	123 ps	104 ps
25	211 ps	107 ps
<20	>150 ps	>230 ps

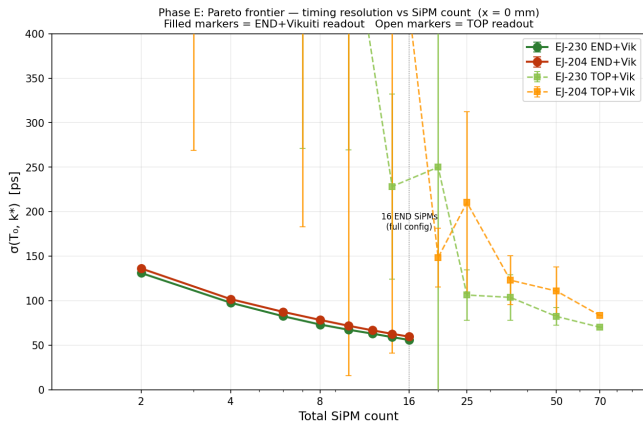
Large uncertainty below N=25:

resolution depends critically on *which* SiPMs are selected relative to the muon track.

Dotted lines: END+Vik reference (56/60 ps).

TOP needs $\gtrsim 50$ SiPMs to match END+Vik resolution.

Phase E: Pareto frontier



END+Vikuiti dominates the Pareto frontier at every SiPM count.

At the same total SiPM budget:

- 4 END SiPMs → 98 ps
vs. 4 TOP SiPMs → >1000 ps
- 8 END SiPMs → 73 ps
vs. 70 TOP SiPMs → 70 ps
- 16 END SiPMs → 56 ps
vs. 70 TOP SiPMs → 70 ps

END+Vik with 16 SiPMs beats TOP with 70 SiPMs by 20%.

⇒ Reflector on end faces is the most cost-effective improvement per SiPM.

What genuinely requires new Geant4 simulation

Computable from existing data:

- k -scan and m -average, any $k, m \leq N_{\text{pe}}$
- END software ablation (8 to 1 SiPM/face)
- TOP software ablation (70 to 1 SiPMs)
- Position dependence of all above
- MC-truth bias $\langle T_0 - T_0^{\text{true}} \rangle$ vs. x

NOT computable from existing data:

- 1 **END+Vik + sparse TOP simultaneously**
In `scan_resolution`: top SiPMs absorb photons $\Rightarrow$ end hits $\approx 1/\text{ev}$.
Need combined geometry: Vikuiti at ends + $\mathcal{O}(\text{few})$ top SiPMs.
- 2 **Physical SiPM reduction ($< 8/\text{end}$):**
Software masking changes information, not optical boundaries.
True optical effect of removing SiPMs from face requires new sim.
- 3 **EJ-200 position scan with reflector:**
Only `scan_end_vikuiti` has EJ-200; additional systematic checks possible.

Proposed minimal new campaign:

Geometry	Scint.	Goal
END+Vik + 4 TOP	EJ-230	Combined T_0
END+Vik + 8 TOP	EJ-230	Combined T_0
4 SiPMs/end (phys.)	EJ-230	True ablation
2 SiPMs/end (phys.)	EJ-230	Minimum config

≈ 4 new geometries $\times$ 13 positions $\times$ 5000 ev

≈ 4 hours total runtime on t0minidaq.

Can reuse run-script infrastructure.

Best configuration from existing data:

Configuration	SiPMs	$\sigma(T_0)$
EJ-230 + END + Vikuiti + $m^*=7$	16	50.5 ps
EJ-230 + END + Vikuiti + $k^*=2$	16	56.0 ps
EJ-204 + END + Vikuiti + $m^*=10$	16	53.4 ps
EJ-230 + TOP + TIR, $k=1$	70	70.2 ps
EJ-204 + TOP + TIR, $k=1$	70	82.1 ps
EJ-230 + END + Vik, 2/end, k^*	4	97.9 ps

Main results:

- 1 **END+Vikuiti beats TOP** at equal SiPM cost by a factor of $\approx 2-5$.
- 2 **m -average $\sim 10\%$ better** than single k -th photon; optimal $m^* = 5-12$.
- 3 **EJ-230 best at centre**; EJ-200 competes at extremes (longer L_{att}).
- 4 **4 END SiPMs total** $\rightarrow < 100$ ps (software ablation).
- 5 **TOP needs $\gtrsim 50$ SiPMs** for comparable timing.
- 6 **Combination** END+Vik + sparse TOP requires new simulation.

Recommendation: run EJ-230 + END+Vik with $m^* = 7$ as baseline; explore physical 4-SiPM/end geometry with new Geant4 campaign.